

ASSIGNMENT #1

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
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Due Date: Wednesday January 22 by 5:00pm in my office

Number of points.....100

Your score..... 89

Where x_i is the mole fraction of each component in the product stream calculated from Table 2

γ_i is the liquid activity coefficient of each component

ν_i is the stoichiometric coefficient of each component

$$K_{eq} = \frac{x_{Acetic\ acid} \gamma_{Acetic\ acid} x_{Ethanol} \gamma_{Ethanol}}{x_{Ethyl\ acetate} \gamma_{Ethyl\ acetate} x_{Water} \gamma_{Water}} = 15.17$$

Discussed?

Table 3: Liquid Activity Coefficients

Liquid Activity Coefficients	Value
Acetic Acid	0.621628
Ethanol	1.34067
Water	2.06848
Ethyl Acetate	1.68614

Calculating Water Flow Rate

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$$Q_{cool} = F_w C_{p_w} (T_w - T_c)$$

Where Q_{cool} is the heat released from the reactor, F_w is the flow rate of water, C_{p_w} is the heat capacity of water

T_w is the temperature of the cooling water entering the jacket around the reactorⁱⁱ

T_c is the temperature of the cooling water leaving the jacket

$$F_w = \frac{Q_{cool}}{C_{p_w} (T_w - T_c)} = \frac{-2.04 \times 10^7 \frac{J}{min}}{75400 \frac{J}{kmol \cdot K} (305.37 K - 343.15 K)} = 15.04 \frac{kmol}{min}$$

Calculating Overall Heat Transfer Coefficient (UA)

As there is heat transfer from the reactor to the cooling water, the following equation is used to find the overall heat transfer coefficient.

$$UA = \frac{Q}{(T_r - T_c)} = \frac{2.04 \times 10^7 \frac{J}{min}}{(343.15 K - 323.15 K)} = 1020405 \frac{J}{min \cdot K}$$

Where Q is the heat duty going into the cooling water and T_r is the temperature of the reactor (70°C)

Table 4: Parameter Results

Parameter	Unit	Result
Feed Density	kmol/m ³	17.8
Product Density	kmol/m ³	16.66
Feed Heat Capacity	J/kmol-K	1.00E+05
Product Heat Capacity	J/kmol-K	1.33E+06
Heat of Reaction	J/kmol	-1.51E+07
Equilibrium Constant		15.17
Water Flow Rate	kmol/min	15.22
Overall Heat Transfer Coefficient	J/min-K	1.02E+06

ⁱⁱ Approximated by usual cooling water utility temperature of 90 degrees Fahrenheit

ⁱⁱⁱ Approximated as this is the highest temperature of cooling water that cooling water towers can handle, 120 degrees Fahrenheit

Table 6: Variables Used

Variable	Symbol	Unit
Concentration of ethanol in the product	CETOH	mol/m ³
Concentration of acetic acid in the product	CAA	mol/m ³
Concentration of ethyl acetate in the product	CEAC	mol/m ³
Concentration of water in the product	CWATER	mol/m ³
Temperature of the Reactor	T _{emp}	K
Kinetic forward rate constant	k _f	mol/m ³ ·min
Kinetic backward rate constant	k _r	mol/m ³ ·min

Calculating Equimolar Initial Mixture

Given the densities and molar masses of acetic acid and water, Excel was used to goal seek the equimolar mixture of water and acetic acid to amount to the 0.5 m³ volume of the reactor. The moles of water and acetic acid came to be 6629.05 moles. The initial concentrations of water and acetic acid were calculated by dividing the moles by their respective volume found in the reactor.

Athena Code

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!Variable Declarations and Data
!*****

Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

Global Tfeed, FAAfeed, FETOHfeed, Fwaterfeed, FEACfeed, Tc As Real
Tfeed=298.15 !Temperature of the feed [Kelvin]
FAAfeed=3216.67 !Flowrate of acetic acid in the feed [mol/min]
FETOHfeed=3116.67 !Flowrate of ethanol in the feed [mol/min]
Fwaterfeed=166.67 !Flowrate of water in the feed [mol/min]
FEACfeed=0
Tc=323.15 !Temperature of the cooling water coming from the jacket [Kelvin]

Global kfo, kro, Ef, Er As Real
kfo=1.14e13 !internal kinetic forward rate constant [mol/m^3*min]
kro=3e12 !internal kinetic backward rate constant [mol/m^3*min]
Ef=59500 !Forward Activation Energy of the reaction [J/mol]
Er=Ef !Backward Activation Energy of the reaction [J/mol]

Global P, Cp, Q, V, Hrxn, UA As Real
P=17797.9; !Molar Density [mol/m^3]
Cp=100.032 !Heat Capacity [J/mol*K]
Q=0.365211 !Volumetric Flowrate [m^3/min]
V=0.5 !Constant Volume of the Reactor [m^3]
Hrxn =-1.51e4 !Heat of Reaction [J/mol]
UA=1020405 !Overall Heat Transfer Coefficient [J/min*K]

@Initial Conditions
U(1)=0; !Initial ethanol conc in reactor [mol/m^3]
U(2)=17484.85; !Initial acetic acid conc in reactor [mol/m^3]
U(3)=0; !Initial ethyl acetate conc in reactor [mol/m^3]
U(4)=54845.11; !Initial water conc. in reactor [mol/m^3]
U(5)=323.15; !Initial temperature of the reactor [K]
  
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Aspen Graphs

